

SHILPA – INCLUSIVE MOBILE LEARNING APP FOR VISUAL, HEARING, PHYSICAL AND COGNETIVELY IMPAIRED STUDENTS (GRADE 3 - 5)

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Specializing in Information Technology

Department of Information Technology

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(Dissertation submitted in partial fulfillment of the requirements for the Bachelor of
Science Special (honors) in Information Technology, Specializing in Information
Technology)

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor:

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Date

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Signature of the co-supervisor:

.....
Date

ABSTRACT

Physical disabilities present significant barriers for primary school students in grades 3-5, limiting their ability to interact with traditional touch-based educational applications and hindering independent learning. Existing assistive technologies often focus on single input methods and lack comprehensive solutions tailored for young learners with diverse physical impairments. To address this gap, this research proposes an inclusive mobile learning application developed with Flutter that integrates multiple adaptive interaction methods including voice-controlled navigation, dwell-time activation, gaze tracking, and facial gesture recognition. The system incorporates customizable accessibility features such as adjustable button sizes, dwell time duration, and visual feedback mechanisms to accommodate individual motor abilities and preferences. Safety features including emergency voice commands, visual countdown indicators, and accidental touch cancellation ensure secure and frustration-free interaction for young users. Additionally, the application provides an AI-powered assistant for learning support, achievement-based motivation systems with collectibles, and real-time progress tracking to enhance engagement and educational outcomes. Cross-platform compatibility ensures accessibility across Android and iOS devices. By combining multi-modal input methods, personalized customization options, and child-friendly design principles, the proposed solution delivers a comprehensive and empowering educational platform that enables students with physical disabilities to access digital learning content independently, safely, and effectively.

Keywords: Physical disabilities, Inclusive education, Multi-modal interaction, Voice control, Facial gesture recognition

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AT	Assistive Technology
API	Application Programming Interface
AI	Artificial Intelligence
iOS	iPhone Operating System
WCAG	Web Content Accessibility Guidelines
AWS	Amazon Web Services
UI/UX	User Interface/User Experience
BSc	Bachelor of Science

1. INTRODUCTION

In today's digital learning environment, accessibility is not just a feature, it is a necessity. Students with physical disabilities often face significant challenges in navigating and interacting with traditional mobile applications, which can limit their ability to learn effectively and independently. My part of the project focuses on designing and implementing an inclusive interface within our e-learning mobile app, specifically tailored for students with physical impairments in grades 3-5.

Students with physical disabilities in this age group represent a particularly vulnerable population in the digital divide. At this critical developmental stage, children are building foundational academic skills while developing their sense of independence. Existing assistive technologies, while valuable, often suffer from critical limitations that prevent them from serving as comprehensive solutions. Most current solutions focus on addressing single interaction modalities such as switch access or voice commands without considering the complex needs of students who may benefit from multiple alternative input methods. Furthermore, many assistive technologies are designed for adult users, lacking the child-friendly interfaces and age-appropriate content that younger learners require.

This section of the application will integrate multiple adaptive interaction methods to ensure that students with limited hand or finger mobility can still engage with lessons, complete activities, and navigate the platform with ease. Core functionalities such as one-switch and voice-controlled navigation, large touch-friendly buttons, and Bluetooth assistive device compatibility will be combined with advanced features like dwell-time activation, gaze tracking, tilt-based navigation, and facial gesture controls. These features will allow students to interact using various body parts or alternative input methods, reducing dependency on fine motor skills and providing multiple pathways for successful interaction.

Additionally, the interface will support customizable settings such as dwell time duration, button size, and feedback type to adapt to the unique needs of each learner. Students with physical disabilities represent a highly diverse group with varying degrees of motor control and preferred interaction methods, making personalization essential for effective learning. Safety and error prevention are also prioritized through visual countdown indicators, the ability to

cancel accidental touches, and an emergency voice or sound-based stop command to ensure secure and frustration-free interaction for young users.

The technical implementation leverages cutting-edge technologies including Flutter for cross-platform development, ensuring compatibility across Android and iOS devices. Advanced AI models from Hugging Face enhance speech recognition accuracy for children's voices, improve gaze estimation precision, and enable sophisticated facial gesture recognition. Real-time data processing provides immediate feedback and accurate grading for student activities, while achievement-based motivation systems with collectible rewards encourage continued engagement with subject-related activities.

By blending practicality with innovative control methods, this part of the project aims to create a truly accessible, comfortable, and empowering learning experience for students with physical disabilities. The integration of a friendly AI assistant provides contextual help and support, reducing dependence on external assistance and promoting independent learning. Through this comprehensive approach, the application will demonstrate how multiple assistive technologies can be seamlessly integrated into mainstream educational platforms, ultimately creating a learning environment where physical disabilities do not determine educational outcomes.

1.1 Background & Literature Survey

Inclusive education is a fundamental right, yet students with physical disabilities in Grades 3–5 often face significant barriers in accessing digital learning tools. Many existing educational applications lack adaptive interfaces tailored to their unique needs, such as limited motor control or reliance on assistive devices. Existing studies primarily focus on assistive technologies for individual disability types, but minimal research has addressed comprehensive, child-friendly solutions that integrate multiple accessibility features into a single platform. This research aims to bridge that gap by developing a mobile application module specifically designed for physically impaired students, incorporating voice commands, large interactive buttons, and Bluetooth assistive device compatibility to enhance accessibility and engagement.

Most learning apps for students with disabilities only help with one problem at a time like apps just for blind students or just for deaf students. They often don't work well for young kids, don't support local languages like Sinhala, and miss helpful tools like eye-tracking or touch-free controls. This makes learning harder for students with physical disabilities. Our app fixes these problems by combining easy-to-use features in one place.

- **Borgestig, Rytterström & Hemmingsson (2016) – Eye Gaze Performance**

The longitudinal analysis confirms the feasibility of gaze-based AT for children with severe impairments. However, it focuses solely on eye gaze. Unlike this project, it does not integrate multiple adaptive controls, curriculum alignment, or customizable dwell-time and child-specific safeguards.

- **Borgestig et al. (2017) – Gaze-Based AT in Daily Activities (Intervention_Study)**

Shows positive impact of gaze-based technology in everyday tasks but limits scope to functional independence. The proposed project extends this by targeting educational outcomes with multimodal input, motivational rewards, and real-time learning support.

- **Borgestig, Rytterström & Hemmingsson (2017) – Parents’ Experiences**

Highlights parental observations of gaze AT in daily use, noting usability and empowerment benefits. Yet it lacks structured educational frameworks, safety/error prevention, and multimodal pathways. The project addresses these omissions by embedding child-centered learning workflows and protective mechanisms.

- **Hsieh et al. (2021) – Communicative Interaction with and Without Eye-Gaze**

Focuses on how gaze technology affects partner communication among children with complex needs. While informative for interaction, it does not address structured learning, gamification, or multimodal accessibility. The proposed app bridges this by applying gaze in formal curriculum-aligned education.

- **Hsieh (2024) – Feasibility of Eye-Gaze Intervention**

Assesses feasibility of eye-gaze interventions in schools, showing promise for students with severe motor and communication difficulties. However, it remains limited to gaze. The project advances feasibility into a full educational system by layering multiple alternative inputs and customizable features.

- **Wilson et al. (2022) – Eye-Gaze or Switch for Cognitive Testing**

Validates acceptability of gaze/switch-based controls for assessments in cerebral palsy. While important, it remains assessment focused. The proposed system expands to continuous educational engagement with multiple input methods, feedback, and progression tools.

- **Andreassen et al. (2024) – Psychosocial Impact of Eye-Gaze Technology**

Finds positive psychosocial outcomes for children and adults using gaze AT. Unlike this project, it does not connect psychosocial benefits with formalized, safe, child-specific learning modules. The current solution unifies psychosocial empowerment with academic performance and adaptive navigation.

- **Hemmingsson et al. (2018) – Case Report: Eye-Gaze Control in Spinal Injury**

Demonstrates early intervention benefits with gaze control in a single child. The scope is narrow and lacks system-level generalization. The proposed app generalizes these benefits into a scalable, child-friendly educational platform with multimodal and localized support.

- **Masayko et al. (2024) – Parents’ Perceptions of Eye-Gaze AAC**

Explores family perceptions of AAC through gaze technology. While emphasizing usability, it does not incorporate multiple modalities or structured pedagogy. This project embeds parental/teacher tracking within a multimodal, curriculum-aligned framework.

- **Kairy et al. (2021) – Mobile App for Social Participation**

Presents a mobile app for physical disability support, validated for usability, but its focus is on social participation. Unlike this project, it lacks multimodal navigation, child-friendly learning design, and localized content.

- **Mott et al. (2018) – Smartphone Photography Accessibility**

Analyzes accessibility barriers for motor-impaired users in photography. Insights highlight UI limitations but do not target education or multimodal pathways. The proposed solution applies similar accessibility learnings but extends them into multimodal educational interaction.

- **Cicek et al. (2020) – Head-Based Pointing on Smartphones**
Develops and evaluates head-based pointing as an input method. While relevant to alternative access, it does not integrate multiple input streams, safety layers, or educational content. This project expands head/face-based gestures into a broader multimodal, child-centered learning system.
- **Fan et al. (2020) – Eyelid Gestures on Mobile Devices**
Proposes eyelid gesture interactions for motor-impaired users. Innovative but narrow in scope, with no child-oriented safety features or educational context. The project incorporates facial gestures among other modalities, adapted for children’s learning tasks.
- **Castellucci& MacKenzie (2019) – TiltWriter**
Explores tilt-based text entry without touch. While valuable, it is restricted to text input. The proposed solution adapts tilt into a multimodal educational framework with gamification and adaptive UI.
- **Ljubic et al. (2014) – Predicting Tilt-Based Input**
Presents models of tilt-based input speed. Useful for benchmarking but not implemented in educational systems. The proposed app transforms tilt into one of several usable child-centered interaction methods.
- **Wobbrock & Myers (2006) – Trackball Text Entry**
Early innovation in motor accessibility through trackball input. However, limited to text and adult usability. This project instead provides multimodal flexibility integrated with learning goals and child safety.
- **Malu et al. (2018) – Accessible Smartwatch Interactions**
Explores smartwatch interactions for motor impairments. While broadening device form factors, it is not aligned with education. This project situates alternative inputs in a structured educational platform for children.

- **Chan et al. (2022) – Eye-Tracking Training Improves Learning**
Finds eye-tracking can improve memory and learning outcomes. Strongly relevant but limited to training studies. The proposed app integrates gaze tracking into multimodal educational workflows with motivational and safety layers.
- **Rytterström et al. (2016) – Teachers’ Experiences with Eye-Gaze Computers**
Captures teachers’ experiences of gaze-controlled learning tools. Valuable insights into classroom acceptance but limited to single modality. This project broadens input diversity and introduces customizable, localized design for Sri Lankan primary learners.
- **Jayasekara et al. (2024) – Smart Wheelchair Robot + Mobile App (Sri Lanka)**
Demonstrates Sri Lankan innovation in accessibility for mobility assistance and home management. While impactful, it does not address educational inclusion. The proposed app fills this gap by applying locally relevant assistive technologies directly to learning environments.

1.2 Research gap

Inclusive education emphasizes equal opportunities for all learners, yet children with physical disabilities in primary education (Grades 3–5) continue to face significant challenges when accessing digital learning platforms. Many existing educational applications provide only limited accessibility, with most designed for a single type of disability such as visual or hearing impairments. While these tools are valuable, they often fail to address the complex and diverse needs of children with physical disabilities who may rely on multiple assistive devices and require adaptable, child-friendly interfaces. Furthermore, current solutions rarely incorporate local language support, making it even harder for children in non-English-speaking regions, such as Sri Lanka, to access and benefit from them.

A review of existing literature highlights these gaps clearly. Paulus and Remijn (2021) provide controlled evidence that dwell-time activation is a feasible alternative to manual input, yet this work is confined to performance testing and does not extend to real-world educational systems or safety-enhanced, child-focused design. Similarly, Borgestig and colleagues across multiple studies (2016; 2017 intervention; 2017 parental experiences) demonstrate that gaze-based assistive technology can support children's independence and daily activities. However, these contributions remain focused exclusively on eye-gaze as a single modality, leaving out the integration of complementary controls such as dwell-time customization or voice input, and overlooking the importance of multimodal choice for children with varying motor abilities.

Expanding on these findings, Hsieh et al. (2021) and Hsieh (2024) examine communicative interaction and school-based feasibility of gaze interventions, confirming that gaze can facilitate participation but again limiting exploration to one input pathway. Wilson et al. (2022) validate the acceptability of gaze and switch controls for cognitive testing among children with cerebral palsy, but their work stops short of embedding these methods into a multimodal educational framework. In a related way, Andreassen et al. (2024) and Masayko et al. (2024) capture the

psychosocial impact and parental perceptions of gaze-based assistive technology, offering valuable perspectives on acceptance and quality of life. Still, these studies do not propose mechanisms for integrating voice and dwell alongside gaze to reduce fatigue and increase flexibility for learners. Hemmingsson et al. (2018) illustrate the benefits of early gaze-based intervention for a child with spinal cord injury, but the single-case nature and gaze-only focus again underscore the lack of multimodal strategies.

Other strands of work further highlight fragmentation. Kairy et al. (2021) presents a mobile app to support social participation for people with disabilities, but it is not designed around adaptive learning contexts or multimodal control. Mott et al. (2018) investigate smartphone photography accessibility for motor-impaired users, identifying UI barriers but not addressing learning or child-specific multimodal interaction. Likewise, Cicek et al. (2020) and Fan et al. (2020) propose head-based pointing and eyelid gestures, which innovate on alternative controls but remain isolated technologies without integration into voice or dwell-time frameworks suitable for primary education. Studies on tilt input, such as Castellucci and MacKenzie (2019) and Ljubic et al. (2014), provide insights into no-touch control, but tilt has limited applicability in structured learning for children and is not considered further in this component. Early work like Wobbrock and Myers (2006) on trackball text entry and Malu et al. (2018) on smartwatch interactions for motor-impaired users reinforce the same limitation: creative input methods remain siloed rather than unified in a multimodal, safety-aware, educational platform.

At the same time, research has begun to connect eye-tracking with cognitive and learning benefits. Chan et al. (2022) show that eye-tracking training can improve learning and memory outcomes, confirming the educational potential of gaze input. Yet, when implemented in isolation, gaze risks over-reliance and fatigue, highlighting the need to combine it with voice and dwell-time activation. Teacher perspectives also reflect this gap: Rytterström et al. (2016) document the benefits of gaze-controlled computers in classrooms, but stress that adoption is shaped by ease of use and

contextual fit needs that can be better addressed through multimodal, customizable interfaces.

Finally, Sri Lankan work by Jayasekara et al. (2024) demonstrates strong local innovation in accessibility through a smart wheelchair and mobile application. While significant in mobility and home management, it does not address inclusive learning or multimodal interaction for children in primary schools. This highlights a gap between technological development and localized educational inclusion.

Taken together, these studies show that while dwell-time, gaze, and alternative input technologies have each been validated in isolation, there remains a critical absence of integrated, multimodal, child-centered systems. Existing solutions either focus narrowly on one modality, overlook the necessity of safety and error-prevention features, or fail to adapt to the needs of young learners in non-English-speaking contexts. This project addresses these limitations by positioning dwell-time activation, gaze tracking, and voice input as complementary modalities, ensuring that accessibility is not only technically possible but also safe, engaging, and developmentally appropriate for physically disabled students in Grades 3–5.

Features Research	Dwell Time	Voice Active	Gaze Track	Multi Option	Child Safety UI	Localization
A Mobile App to Optimize Social Participation [11].						
Longitudinal eye-gaze performance [2].						
Eye-Gaze Control Technology as Early Intervention [9].						
photography accessibility (motor impairments) [12].						
Smart Wheelchair Robot and Mobile Application [21].					Partial	
Our Proposed System						

Table 1: Comparison of Research Gaps in existing studies and Proposed Mobile App

1.3. Research problems

Despite advancements in assistive technology, students with physical disabilities in Grades 3-5 continue to face significant barriers in accessing digital learning tools.

Existing solutions suffer from critical limitations:

- **Can fragmented accessibility features be overcome by integrating multiple methods into a single system?**

Most assistive technologies rely on single input methods (e.g., gaze control, or voice commands) without integrating multi-modal alternatives. This restricts flexibility, forcing users to depend on one interaction mode even when it is inefficient or fatiguing.

- **How can assistive technologies be better adapted to meet the needs of young learners?**

Voice recognition systems are primarily trained on adult speech, leading to inaccuracies for children. Additionally, many assistive tools lack child-friendly interfaces, adjustable settings (e.g., dwell time, button size), and support for local languages.

- **What safety and error-prevention features are necessary to make e-learning tools reliable for physically impaired students?**

Current solutions neglect essential safety features, such as emergency stop commands, accidental input cancellation, or visual feedback mechanisms, increasing frustration and hindering independent use.

- **Is it possible to create a unified educational solution that combines adaptive navigation with curriculum-aligned learning features for young students?**

While some apps address individual disabilities (e.g., switch access for motor impairments or screen readers for visual impairments), none combine adaptive navigation (voice, gaze, facial gestures) with learning-specific needs such as auto-play lessons, simplified UI, and curriculum-aligned content for early education.

This research seeks to bridge these gaps by developing a comprehensive, child-friendly e-learning interface that:

- Integrates multi-modal input (voice, gaze, switches, facial gestures) to reduce dependency on a single control method.
- Prioritizes young learners through age-appropriate voice commands, adjustable interaction settings, and localized language support.
- Ensures safety and usability with emergency stop functions, error cancellation, and visual feedback.
- Supports personalized learning with customizable dwell times, button sizes, and adaptive navigation tailored to individual motor abilities.

1.4. Research Objectives

1.4.1 Main Objective

The main objective of this research was to design and develop an adaptive e-learning mobile interface that enables students with physical impairments to access and interact with learning content independently, safely, and efficiently. The system was designed to reduce the dependency on traditional touch-based interaction by integrating multiple alternative input methods such as voice commands, gaze-based navigation, and adaptive dwell-time selection.

In addition, the solution aimed to provide a personalized and user-friendly learning experience by allowing customization of interaction settings such as dwell time and interface elements. The overall goal was to improve accessibility, enhance user engagement, and support independent learning for students with limited motor control.

1.4.2 Specific Objectives

- To integrate multiple interaction methods such as voice control, gaze-based navigation, and dwell-time activation to support users with limited motor abilities.
- To design and implement an adaptive dwell-time mechanism that adjusts interaction duration based on user performance to improve selection accuracy and reduce errors.
- To develop a gaze-based navigation system using facial and eye tracking techniques to enable hands-free interaction with the application.
- To implement voice command functionality that allows users to perform navigation and system control tasks efficiently without physical touch.
- To create a user-friendly and customizable interface by allowing adjustments to settings such as dwell time and interaction feedback, ensuring suitability for different user needs.

- To evaluate the effectiveness of different interaction modes (voice-only, gaze-only, dwell-time, and hybrid) based on performance metrics such as task completion time, accuracy, and error rate.
- To identify the most effective interaction approach by comparing different input methods and selecting the optimal hybrid interaction model for improved usability and performance.

2. METHODOLOGY

2.1 Research Design and Overall Procedure

The physical impairment support component was designed using a design and evaluate approach to enable hands free and touch free interaction for students with limited motor control. The system was developed as an Adaptive Multimodal Interaction Optimization Framework that combines multiple input methods to improve accessibility and usability.

The implementation focused on three main interaction methods:

- Adaptive Dwell-Time Selection
- Eye/Head Gaze-Based Navigation
- Voice-Based Command Navigation

In addition, a multimodal evaluation process was carried out to measure performance and identify the most effective interaction method.

2.2 System Overview

The system was developed as a multimodal interaction framework that allows users to interact using different input methods based on their ability and comfort. Instead of relying on a single input method, the system combines gaze, voice, and dwell-time to provide flexible and adaptive control. A centralized controller was used to manage all interaction inputs and ensure stable and conflict-free operation between different modes.

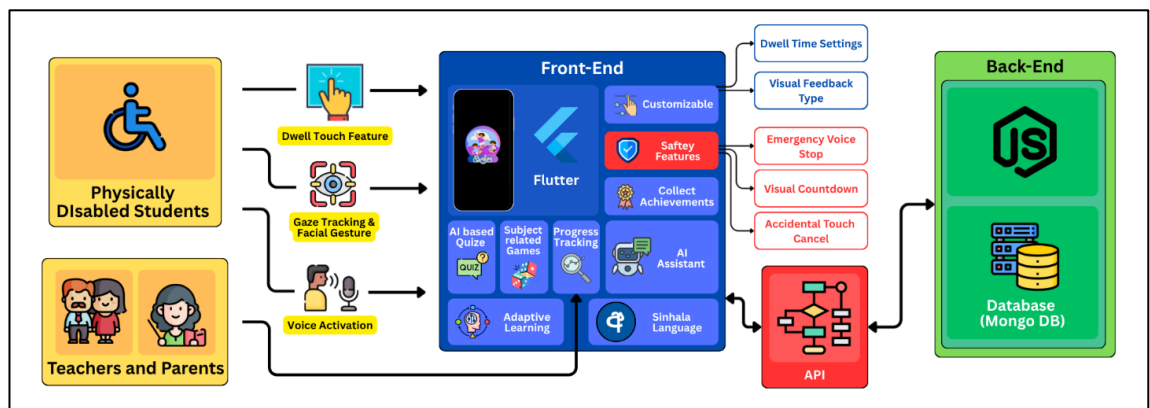


Figure 1: Overview Diagram

2.3 Adaptive Dwell Time Selection

2.3.1 Rational

Traditional dwell-time systems use a fixed time duration (e.g., 2 seconds) to select an item. However, this creates problems such as accidental selections for unstable users and slow interaction for experienced users. To solve this, an adaptive dwell-time mechanism was introduced, which automatically adjusts the selection time based on user behavior.

2.3.2 Base Implementation

The dwell-time mechanism was implemented using Flutter's native components:

- Listener for detecting user focus
- Timer for tracking dwell duration
- AnimationController for visual progress
- CustomPainter for rendering feedback

When a user focuses on a UI element:

1. The system detects interaction
2. A timer starts based on the current dwell duration
3. A visual indicator shows progress
4. If focus is maintained, selection is triggered automatically
5. If focus is removed, the timer resets

This enables fully touch-free selection

2.3.3 Adaptive Dwell Time Algorithm

The dwell-time dynamically adjusts based on user performance:

- Initial dwell-time: 2.0 seconds
- Minimum limit: 1.0 second
- Maximum limit: 3.0 seconds

The system monitors:

- Reaction time
- False selections
- Cancellation attempts
- Task completion success

Adjustment rules:

- If the user successfully selects items multiple times → decrease dwell-time (faster interaction)
- If the user frequently cancels or makes errors → increase dwell-time (more stability)

For example:

- 5 successful selections → reduce time by 0.1 seconds
- 3 cancellations → increase time by 0.2 seconds

This creates a personalized and self-learning interaction system that adapts over time.

2.4 Eye and Head Gaze Based Navigation

2.4.1 Gaze Detection Mechanism

Gaze-based navigation was implemented using facial landmark detection. The system extracted:

- Eye position
- Nose position
- Head orientation (yaw, pitch, roll)

These values were mapped to screen coordinates to identify where the user is looking.

2.4.2 Gaze Stability Filtering

Raw gaze data can be unstable due to small movements and camera noise. To improve accuracy, a stability filtering mechanism was applied:

- Smoothing using moving average / EMA
- Spike detection to remove sudden jumps
- Stability window to confirm user intention

A selection is triggered only when:

- Gaze remains within a target area
- Movement is stable for a short duration

This prevents accidental selections and improves control accuracy.

2.4.3 Calibration Step

A calibration phase was introduced to adapt the system for each user:

1. User looks at predefined points
2. System measures natural eye movement (variance)
3. Personalized thresholds are generated

This ensures the system works accurately for different users.

2.4.4 Blink based Selection

A blink-based interaction method was introduced as an alternative to dwell-time:

- The system detects eye closure probability
- If a blink occurs while focusing on a target, selection is triggered instantly

This reduces waiting time and improves interaction speed.

2.5 Voice Based Navigation

2.5.1 System Design

Voice interaction was implemented as a hands-free alternative for users who cannot rely on gaze stability. The system includes:

- Speech-to-text conversion
- Command mapping
- Intent detection

2.5.2 Interaction Workflow

The voice interaction follows these steps:

1. User activates voice mode
2. Speech input is captured
3. Audio is converted to text
4. The system matches commands such as:
 - “Next”, “Back”, “Select”, “Open”
5. The corresponding action is executed

Invalid commands are ignored to prevent errors.

2.5.3 Advanced Features

To improve usability:

- Fuzzy matching is used to understand similar words
- Sinhala and English commands are supported
- Users can select items using numbers (e.g., “Select 1”)

This improves flexibility and ease of use.

2.6 Multimodal Interaction

The system supports four interaction modes:

1. Voice Only
2. Gaze Only
3. Gaze with Adaptive Dwell-Time
4. Hybrid Mode (Voice + Gaze)

In hybrid mode:

- Gaze acts as a pointer
- Voice acts as the selection trigger

This allows faster and more natural interaction compared to using a single method.

A centralized multimodal controller ensures:

- No conflict between inputs
- Stable interaction behavior
- Priority for confirmed user actions

2.2.1 Methodology

The physical impairment support component was developed using a design-and-evaluate approach to enable hands-free and touch-free interaction for learners with motor control limitations. The system was designed as an Adaptive Multimodal Interaction Optimization Framework that integrates multiple interaction methods to improve usability and accessibility. The implementation focused on three main interaction techniques, namely adaptive dwell-time selection, gaze-based navigation, and voice-based command interaction, followed by a structured evaluation process to measure system performance across different interaction modes.

The adaptive dwell-time mechanism was introduced to overcome the limitations of fixed dwell durations. Instead of using constant selection time, the system dynamically adjusted the dwell-time based on user interaction behavior. The implementation was carried out using Flutter's native components such as Listener, Timer, AnimationController, and CustomPainter to ensure accurate event handling and real-time feedback. When a user focused on a UI element, the system initiated a timer and displayed a visual progress indicator. If the user maintained focus, the selection was automatically triggered, while loss of focus resulted in cancellation. The dwell-time was continuously optimized by monitoring performance metrics such as reaction time, false selections, cancellation attempts, and task success. Based on these observations, the system increased dwell-time when instability was detected and reduced it when users demonstrated consistent accuracy, maintaining the duration within predefined limits.

Gaze-based navigation was implemented using facial landmark detection to track eye movement and head orientation. The system extracted key features such as eye position, nose coordinates, and head pose angles, and mapped them to screen coordinates to determine user intent. Since raw gaze data can be unstable, a gaze stability filtering mechanism was applied to improve accuracy. This included smoothing techniques to reduce noise, spike detection to eliminate sudden movements, and a stability window that ensured selections were only triggered when the gaze remained consistent within a target area for a specific duration. Additionally, a

calibration process was introduced to personalize the system for each user by measuring natural eye movement patterns and generating customized stability thresholds.

Voice-based navigation was implemented as an alternative interaction method for users who have trouble maintaining stable gaze or dwell focus. The system captured audio input and converted it into text using a speech-to-text engine. The recognized text was then matched with predefined navigation commands through a command mapping system and intent classification logic. Valid commands triggered corresponding user interface actions, while invalid inputs were ignored to prevent unintended operations. To improve reliability, the system accepted only predefined commands, rejected low-confidence inputs, and ensured that execution occurred only after complete speech detection. Logging mechanisms were also included to monitor recognition accuracy and system performance.

The multimodal interaction framework combined all input methods into a unified system that supported multiple interaction modes, including voice-only, gaze-only, gaze with adaptive dwell-time, and a hybrid mode combining voice and gaze. A centralized controller managed all interaction inputs and ensured stable operation without conflicts between different modalities. In hybrid mode, gaze was used to identify the target, while voice was used to confirm the action, resulting in faster and more natural interaction.

System performance was evaluated by allowing users to perform predefined tasks under different interaction modes. Key performance metrics such as task completion time, selection accuracy, error rate, false activation count, and user feedback were recorded. The results from each mode were compared to analyze improvements in efficiency, accuracy, and interaction stability. This evaluation process enabled the identification of the most effective interaction method and demonstrated the benefits of combining multiple input techniques in a single adaptive framework.

2.2.2 Summary of methodology

Research	Work	Methods & Algorithms to be used	Expected Accuracy	Remarks
Adaptive Multimodal Interaction Framework	Development of a hands-free and touch-free mobile interaction system for students with motor impairments using multiple input methods (gaze, voice, dwell-time)	Adaptive Dwell-Time Algorithm (dynamic adjustment based on user performance), Flutter-based event handling (Listener, Timer, AnimationController), Performance logging	Improved selection accuracy and reduced interaction delay	Provides personalized interaction by adjusting dwell-time dynamically based on user behavior

Table 2: methodology summary table

2.7 High-Level Architecture Diagram

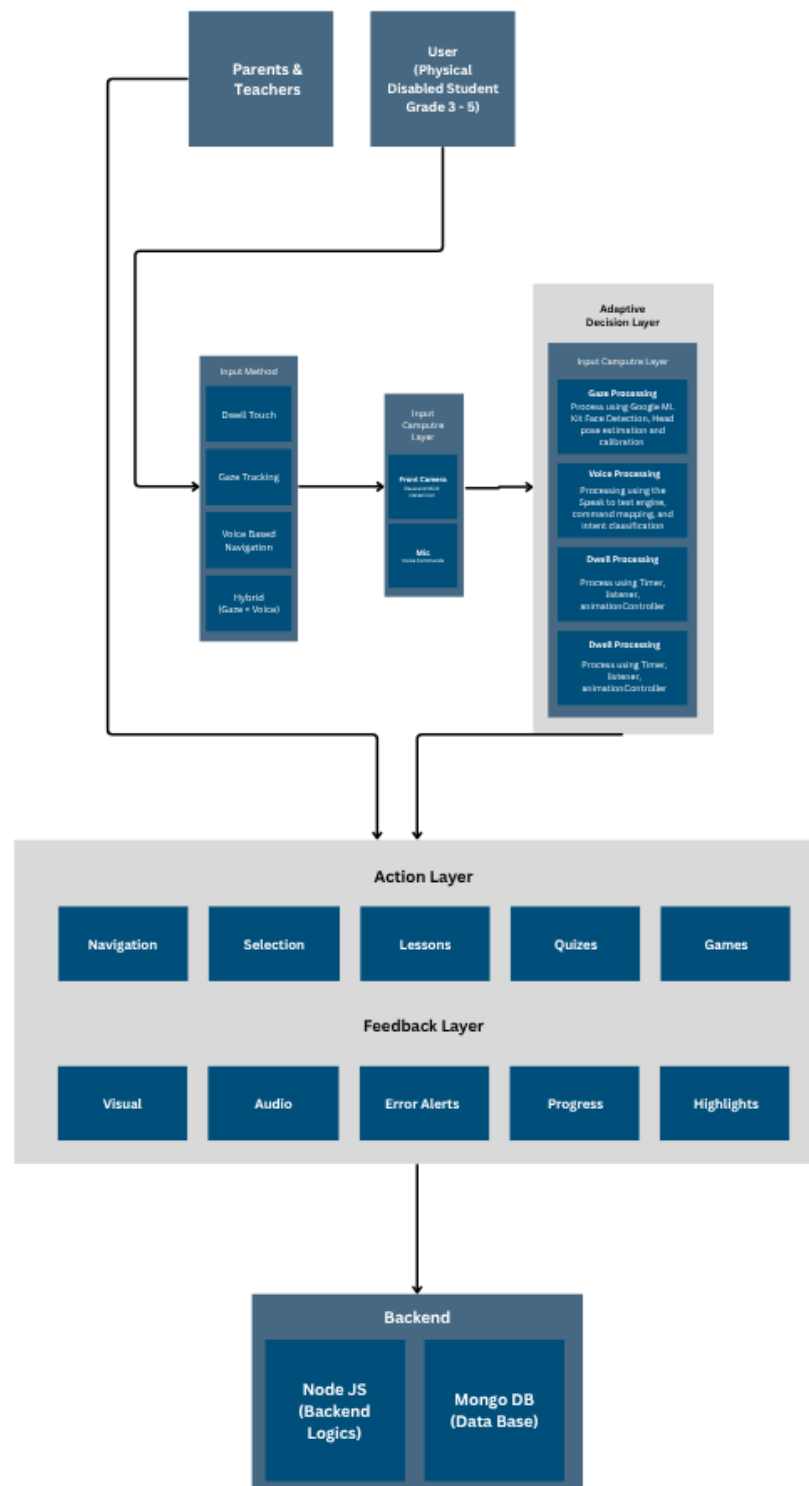


Figure 2: Software Architectural Diagram

2.8 Project Requirements that have been achieved

2.8.1 Functional Requirements

The physical impairment support module successfully achieved the required functional capabilities by implementing multiple interaction methods that allow users to operate the system without relying on traditional touch input.

The system enabled adaptive dwell-time selection, allowing users to select items by maintaining focus on a target. The dwell-time was dynamically adjusted based on user behavior, improving both accuracy and efficiency. Gaze-based navigation was implemented using facial and eye tracking, allowing users to control the interface through head and eye movement. Stability filtering ensured that selections were triggered only when the user's gaze remained stable, reducing accidental activations.

Voice-based navigation was also successfully implemented, enabling users to control the application using predefined commands such as "Next", "Back", and "Select". The system processed voice input through speech-to-text conversion and command mapping to trigger navigation actions. In addition, hybrid interaction combining voice and gaze was achieved, allowing users to select items more quickly by looking at a target and confirming using voice.

The system also included features such as blink-based selection for faster interaction, dynamic input mode switching, and continuous performance logging. These functionalities ensured that users could interact with the system efficiently using different methods based on their abilities.

2.8.2 Non-Functional Requirements

The system met key non-functional requirements related to usability, reliability, performance, and accessibility.

Usability was improved by providing a simple and user-friendly interface with large interactive elements and clear visual feedback. The adaptive dwell-time mechanism and gaze stability filtering enhanced ease of use by reducing user effort and interaction errors.

Reliability was ensured through controlled input processing, where invalid voice commands were ignored and unstable gaze signals were filtered. The system maintained consistent behavior across different interaction modes, ensuring smooth operation even when switching between input methods.

Performance was optimized by implementing efficient algorithms such as smoothing techniques for gaze tracking and optimized event handling. The system maintained low latency in processing user input, ensuring real-time response during interaction.

Accessibility was achieved by supporting multiple input methods including gaze, voice, and dwell-time, allowing users with different levels of motor impairment to interact with the system. Personalization features such as adaptive dwell-time and calibration further improved accessibility by adjusting the system to individual user needs.

2.8.3 Other Requirements

In addition to functional and non-functional requirements, several additional requirements were achieved to enhance the overall effectiveness of the system.

Personalization was a key aspect, where the system adapted to individual users by learning their interaction patterns and adjusting dwell-time accordingly. The calibration process allowed the system to customize gaze sensitivity for each user, improving accuracy.

Safety and error prevention were addressed through features such as stability thresholds, spike detection, and controlled command execution. These mechanisms prevented unintended selections and ensured a safer interaction experience.

The system also supported multilingual voice commands, allowing users to interact using both English and Sinhala, improving usability in the local context. Furthermore, dynamic indexing enabled users to select items using numbers, simplifying navigation.

Continuous performance logging was implemented to record interaction data such as accuracy, errors, and response time. This helped in evaluating system performance and improving interaction efficiency.

Overall, these additional features enhanced the adaptability, usability, and effectiveness of the physical impairment support module.

2.9 Consideration of the aspects of the system

Appropriate coding and development standards were followed during the implementation of the physical impairment support module to ensure code quality and maintainability. Object-oriented principles were used to organize components such as gaze tracking, adaptive dwell-time, and voice-based interaction into structured modules. Important parts of the code were clearly commented, and consistent naming conventions were maintained to improve readability. Performance and reliability were considered by optimizing real-time input handling and including basic error handling for unstable gaze and incorrect voice inputs. In addition, all report writing and referencing were done using the IEEE format to maintain academic standards.

2.9.1 Social aspects

The physical impairment support module contributes to social inclusion by enabling students with limited motor abilities to access digital learning independently. By providing hands-free interaction methods such as gaze, voice, and adaptive dwell-time, the system reduces reliance on assistance from others and promotes equal learning opportunities. This helps reduce the digital divide and allows students to actively participate in educational activities alongside their peers, improving confidence and engagement.

2.9.2 Security aspects

Security aspects were considered to ensure safe and reliable system usage. The module limits input actions to predefined commands in voice interaction to prevent unintended operations. Basic validation and error handling mechanisms were implemented to manage incorrect inputs from gaze or voice. Any user-related data, such as interaction logs, is stored securely and used only for system improvement purposes. These measures help maintain system integrity and protect user interactions from unintended or unsafe behavior.

2.9.3 Ethical aspects

Ethical considerations were addressed by ensuring that the system is designed with fairness, accessibility, and user well-being in mind. The module provides equal access to learning regardless of physical limitations and avoids bias by allowing multiple interaction methods instead of forcing a single approach. User data collected during interaction, such as performance logs, is handled responsibly and only used to improve system functionality. In addition, the system avoids causing user fatigue or frustration by including adaptive mechanisms and safe interaction controls.

2.9.4 Limitations

- The accuracy of gaze-based navigation depends on lighting conditions and camera quality, which can affect performance in real-world environments.
- Small head movements and natural eye jitter may still cause slight instability, even with smoothing and filtering techniques.
- Voice-based interaction may be affected by background noise or unclear speech, leading to incorrect command recognition.
- The system relies on predefined voice commands, which limits flexibility in natural language interaction.
- Adaptive dwell-time requires initial interaction data, so performance may not be optimal during early usage.
- Continuous use of gaze tracking may cause eye fatigue for some users over long periods.
- The system performance may vary across different mobile devices due to hardware limitations such as processing power and camera capabilities.
- Real-time processing of gaze and voice inputs may introduce slight delays on low-end devices.
- The module focuses mainly on upper-limb impairments and may not fully support users with severe multiple disabilities.
- Internet dependency for certain features (e.g. speech recognition) may limit functionality in offline environments.

2.10 Commercialization aspects of the product

2.10.1 Target Audience

- Primary school students (Grades 3–5) with physical impairments or limited motor control
- Students who have difficulty using touch-based mobile applications
- Children who require alternative interaction methods such as gaze, voice, or dwell-time
- Special education students in inclusive or assisted learning environments
- Teachers and educators supporting students with physical disabilities
- Parents or caregivers assisting children with learning at home

2.10.2 Market Space

- Inclusive education technology sector focusing on assistive learning solutions
- Mobile learning applications designed for differently abled students
- Schools and institutions implementing digital learning for special education
- Growing demand for accessible educational tools in developing countries like Sri Lanka
- Lack of integrated multimodal solutions (gaze + voice + dwell) creates a gap in the market
- Opportunity to expand into global accessibility-focused EdTech platforms

2.10.3 Revenue Earning

- Freemium model with basic accessibility features available for free
- Premium subscription for advanced features such as personalized settings and analytics
- Licensing the application to schools and educational institutions
- Collaboration with government or NGOs for inclusive education programs
- In-app educational content or activity-based paid features
- Potential partnerships with assistive technology providers or EdTech companies

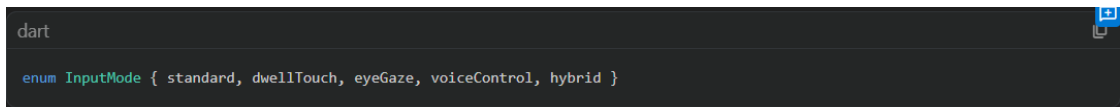
2.11 Testing and Implementation

2.11.1 Code Implementations for the research part

2.11.1.2 Overview and Architecture

The Physical Impairment module of ShilpaMobileLearningApp implements an Adaptive Multimodal Interaction Optimization Framework designed to enable children with physical disabilities to interact with a mobile learning application through alternative input methods. The system provides five interaction modalities. Standard Touch, Adaptive Dwell-to-Touch, Head-Gaze Tracking with Blink-to-Select, Voice-Based Control, and a Hybrid Multimodal Fusion mode and allows users to switch between them at runtime via a floating action button.

The architecture follows a service-oriented pattern built with Flutter (Dart), where each interaction channel is encapsulated in a dedicated singleton service. A centralized InteractionStatusService acts as a global event bus, enabling cross-modal communication. The key enabler of the hybrid fusion approach.



```
dart
enum InputMode { standard, dwellTouch, eyeGaze, voiceControl, hybrid }
```

Figure 3: Enums

This single enum drives the entire framework. Every interactive UI element receives the current InputMode and adapts its behaviour accordingly, making the system fully polymorphic at the interaction layer.

2.11.1.3 Adaptive Dwell Touch

Purpose

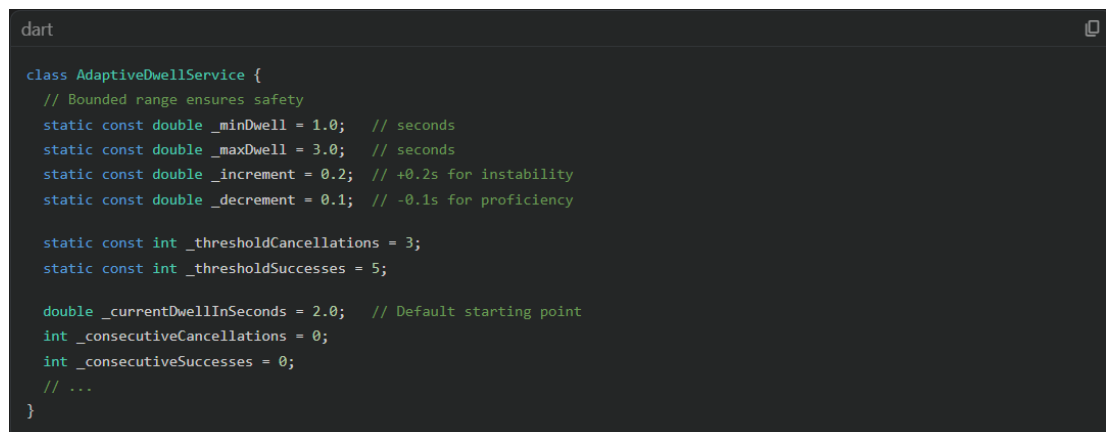
The AdaptiveDwellService dynamically adjusts the dwell activation duration. The time a user must hold a finger on a button before it registers as a tap based on real time interaction performance. This addresses a core challenge for users with motor impairments: a fixed dwell time is either too short (causing accidental selections) or too long (causing fatigue).

Adaptive Algorithm

The algorithm employs a reinforcement-style feedback loop with asymmetric adjustment rates:

- Success path: After 5 consecutive successful selections, the dwell duration is decreased by 0.1 seconds, making the interface faster.
- Cancellation path: After 3 consecutive aborted attempts (lift-off before completion), the dwell duration is increased by 0.2 seconds, stabilizing the interface.

The asymmetry (faster increase than decrease) is deliberate. It prioritizes preventing frustration from false activations over raw speed

A screenshot of a code editor showing the Dart implementation of the AdaptiveDwellService. The code is in a dark-themed editor with a 'dart' label in the top left. It defines a class with static constants for minDwell, maxDwell, increment, and decrement, and static constants for thresholdCancellations and thresholdSuccesses. It also has instance variables for currentDwellInSeconds, consecutiveCancellations, and consecutiveSuccesses.

```
dart

class AdaptiveDwellService {
  // Bounded range ensures safety
  static const double _minDwell = 1.0; // seconds
  static const double _maxDwell = 3.0; // seconds
  static const double _increment = 0.2; // +0.2s for instability
  static const double _decrement = 0.1; // -0.1s for proficiency

  static const int _thresholdCancellations = 3;
  static const int _thresholdSuccesses = 5;

  double _currentDwellInSeconds = 2.0; // Default starting point
  int _consecutiveCancellations = 0;
  int _consecutiveSuccesses = 0;
  // ...
}
```

Figure 4: Adaptive Dwell Mechanism

Success Recording

When a user successfully completes a dwell selection, the service resets the instability counter and increments the proficiency counter. Once the proficiency threshold is

reached and the current duration is above the minimum bound, the system reduces the dwell time.

```
dart

Future<void> recordSuccess() async {
  _consecutiveCancellations = 0;
  _consecutiveSuccesses++;

  if (_consecutiveSuccesses >= _thresholdSuccesses &&
      _currentDwellInSeconds > _minDwell) {
    final oldVal = _currentDwellInSeconds;
    _currentDwellInSeconds = (_currentDwellInSeconds - _decrement)
      .clamp(_minDwell, _maxDwell);
    _consecutiveSuccesses = 0;

    if (oldVal != _currentDwellInSeconds) {
      await _applyAdaptation(oldVal, 'USER_PROFICIENCY_HIGH');
    }
  }
}
```

Figure 5: Dwell Success

Cancellation Recording

Conversely, when a user lifts their finger before the dwell timer completes, it is recorded as an instability signal.

```
dart

Future<void> recordCancellation() async {
  _consecutiveSuccesses = 0;
  _consecutiveCancellations++;

  if (_consecutiveCancellations >= _thresholdCancellations &&
      _currentDwellInSeconds < _maxDwell) {
    final oldVal = _currentDwellInSeconds;
    _currentDwellInSeconds = (_currentDwellInSeconds + _increment)
      .clamp(_minDwell, _maxDwell);
    _consecutiveCancellations = 0;

    if (oldVal != _currentDwellInSeconds) {
      await _applyAdaptation(oldVal, 'USER_INSTABILITY_DETECTED');
    }
  }
}
```

Figure 6: Dwell Process Cancellation

Persistence and Notification

Adapted durations are persisted using SharedPreferences so the system remembers the user's optimal dwell time across sessions. A broadcast Stream<DwellAdaptationEvent> notifies the UI to display real-time feedback (e.g., a SnackBar showing "Interaction Stabilized, New time: 2.4s")

```
dart

Future<void> _applyAdaptation(double oldValue, String reason) async {
  final prefs = await SharedPreferences.getInstance();
  await prefs.setDouble(_keyDwellDuration, _currentDwellInSeconds);

  final event = DwellAdaptationEvent(
    oldValue: oldValue,
    newValue: _currentDwellInSeconds,
    reason: reason,
    direction: _currentDwellInSeconds > oldValue
      ? AdaptationDirection.increased
      : AdaptationDirection.decreased,
  );
  _adaptationController.add(event);
}
```

Figure 7: Applying Adaptation

2.11.1.4 Head Gaze Tracking with Blink to Select Purpose

The EyeTrackingService enables hands-free interaction by tracking the user's head orientation via the front camera and mapping it to an on-screen cursor position. It uses Google ML Kit's Face Detection API to extract facial landmarks and Euler angles (yaw/pitch) in real time at approximately 30 FPS.

Data Model

Each frame produces a GazeData object that encapsulates the estimated screen position, stability status, and blink probability

```
dart

class GazeData {
  final double x;
  final double y;
  final double confidence;
  final bool isStable;
  final double blinkProbability; // 1.0 = eyes open, 0.0 = eyes closed
  final DateTime timestamp;
}
```

Figure 8: Considered Data

Algorithm 1 – Real Time Gaze Stability Filtering

This is the core signal processing pipeline that runs at ~30 FPS. It converts raw, jittery head orientation data into a smooth, stable cursor position through three stages.

Stage 1 - Head Orientation to Screen Mapping:

The face detector provides yaw (left-right) and pitch (up-down) Euler angles. These are linearly mapped to screen coordinates relative to the screen center, with a configurable sensitivity factor.

```
dart

void _processFace(Face face, double imgWidth, double imgHeight) {
  final yaw = face.headEulerAngleY ?? 0;
  final pitch = face.headEulerAngleX ?? 0;

  const sensitivity = 35.0;
  double centerX = size.width / 2;
  double centerY = size.height / 2;

  // Mirrored: +Yaw turns Left on screen
  double targetX = centerX - (yaw * sensitivity);
  double targetY = centerY - (pitch * sensitivity);

  // Apply calibration drift offset
  targetX = targetX + _gazeOffset.dx;
  targetY = targetY + _gazeOffset.dy;

  // Clamp to screen bounds
  targetX = targetX.clamp(0.0, size.width);
  targetY = targetY.clamp(0.0, size.height);
}
```

Figure 9: Screen Mapping Implementation

Stage 2 - Spike Detection and Removal:

A threshold-based filter rejects any frame where the cursor jumps more than 150 pixels from the previous position, eliminating transient detection errors.

```
dart

if (_lastRawPosition != null) {
  double distance = (currentRawPosition - _lastRawPosition!).distance;
  if (distance > _spikeThreshold) return; // Reject outlier frame
}
_lastRawPosition = currentRawPosition;
```

Figure 10: Detect Spikes

Stage 3 - Exponential Moving Average (EMA) Smoothing:

Rather than a simple moving average, the system uses EMA with $\alpha = 0.25$ for superior responsiveness while aggressively suppressing noise.

```
dart

if (_smoothedPosition == null) {
  _smoothedPosition = currentRawPosition;
} else {
  _smoothedPosition = Offset(
    _smoothedPosition!.dx +
    _emaAlpha * (currentRawPosition.dx - _smoothedPosition!.dx),
    _smoothedPosition!.dy +
    _emaAlpha * (currentRawPosition.dy - _smoothedPosition!.dy),
  );
}
```

Figure 11: Smoothing Process

Stage 4 - Stability Window Logic:

The system maintains a rolling history of the last 15 raw positions and calculates variance against the smoothed position. If variance falls below a personalized threshold, the gaze is classified as stable, enabling dwell-based or blink-based selection.

```
dart

bool isStable = false;
if (_history.length >= _historyLimit) {
  double variance = _calculateVariance(_history, smoothedPos);
  isStable = variance < _stabilityVarianceThreshold;
}
```

Figure 12: Stability Logic

Algorithm 2 - Personalized Threshold Generation (Calibration)

During the 5 point calibration process, the system collects gaze samples at known screen positions. After calibration, it computes:

- **Drift Offset** - the systematic error between where the user looks and where the system thinks they look, calculated from the center calibration point.
- **Personalized Stability Threshold** - derived from the user's unique jitter profile (average variance $\times 1.5$, clamped between 15.0 and 60.0 pixels)

```
dart

Future<void> finishCalibration() async {
  // Calculate Drift Offset from center point
  if (_calibrationSamples.containsKey(2)) {
    final samples = _calibrationSamples[2]!;
    double avgX = samples.map((e) => e.dx).reduce((a, b) => a + b) / samples.length;
    double avgY = samples.map((e) => e.dy).reduce((a, b) => a + b) / samples.length;
    _gazeOffset = Offset((size.width / 2) - avgX, (size.height / 2) - avgY);
  }

  // Personalized Stability Threshold
  double avgVar = groups > 0 ? totalVariance / groups : 50.0;
  _calculatedAccuracy = (1.0 - (avgVar / 100.0)).clamp(0.01, 0.99);
  _stabilityVarianceThreshold = (avgVar * 1.5).clamp(15.0, 60.0);
}
```

Figure 13: Calibration Process

This personalization means a child with higher involuntary head movement gets a more lenient stability threshold, while a more controlled user benefits from tighter precision.

Blink-to-Select Mechanism

The blink detection leverages MLKits `leftEyeOpenProbability` and `rightEyeOpenProbability` classifiers. A double blink confirmation protocol is implemented to prevent accidental selection.

```
dart

void _handleBlinkTrigger() {
  if (!_isWaitingForSecondBlink) {
    // FIRST BLINK: Start 2.5-second confirmation window
    setState(() => _isWaitingForSecondBlink = true);
    _interactionService.updateStatus(
      InteractionStatus(state: InteractionState.waitingConfirmation),
    );
    _confirmationTimeout = Timer(const Duration(milliseconds: 2500), () {
      if (mounted && _isGazeInside) {
        setState(() => _isWaitingForSecondBlink = false);
      }
    });
  } else {
    // SECOND BLINK: Confirmed selection
    _confirmationTimeout?.cancel();
    setState(() => _isWaitingForSecondBlink = false);
    if (widget.onTap != null) widget.onTap!();
  }
}
```

Figure 14: Blink to select Implementation

The first blink triggers a "waiting for confirmation" state with visual feedback. If the user blinks again within 2.5 seconds while still gazing at the same element, the selection is confirmed. If not, the system silently resets.

2.11.1.5 Voice Based Control

Speech Recognition Service

The SpeechService wraps the speech_to_text package into a continuous listening loop with automatic restart capability. It initializes with Sinhala (si-LK) locale priority, falling back to the system default.

```
dart

Future<void> init() async {
  var status = await Permission.microphone.status;
  if (!status.isGranted) {
    status = await Permission.microphone.request();
  }

  final available = await _speech.initialize(
    onStatus: (status) {
      if (status == 'done' || status == 'notlistening') {
        _handleListeningStopped();
      }
    },
    onError: (err) {
      if (_islistening) _handleListeningStopped();
    },
  );

  // Sinhala locale prioritization
  final locales = await _speech.locales();
  for (final loc in locales) {
    if (loc.localeId.toLowerCase().startsWith('si')) {
      _localeId = loc.localeId;
      break;
    }
  }
}
```

Figure 15: Speech Recognition Service Implementation

An acoustic confidence threshold filter (≥ 0.5) prevents low-confidence recognitions from triggering false commands.

```
dart

if (result.finalResult && _onCommandRecognized != null) {
  if (confidence > 0.5 || confidence == 0) { // 0 = not reported
    _onCommandRecognized!(result.recognizedWords);
  } else {
    print('Ignored low confidence recognition: $confidence');
  }
}
```

Figure 16: Ignoring Low Confidence Words

Voice Command Parser — Bilingual NLP with Fuzzy Matching

The VoiceCommandParser is the natural language understanding (NLU) engine of the voice subsystem. It implements a multi-strategy parsing pipeline that supports both English and Sinhala commands.

Strategy 1 - Numeric ID Extraction:

(highest priority): Supports both Arabic digits and Sinhala phonetic number words for item selection.

```
dart

static VoiceCommand parse(String text) {
  text = text.toLowerCase().trim();

  // English digit extraction
  final idMatch = RegExp(r'(\d+)').firstMatch(text);
  if (idMatch != null) {
    lastParsedData = int.tryParse(idMatch.group(1)!);
    if (lastParsedData != null) return VoiceCommand.selectById;
  }

  // Sinhala phonetic numbers
  if (text.contains('එක') || text.contains('eka')) {
    lastParsedData = 1;
    return VoiceCommand.selectById;
  }
  if (text.contains('දෙක') || text.contains('deka')) {
    lastParsedData = 2;
    return VoiceCommand.selectById;
  }
  // ... (continues for 3, 4, 5)
}
```

Figure 17: Numbers Support feature

Strategy 2 - Keyword Matching with Fuzzy Fallback:

Each command category maps to a bilingual keyword list. The matcher first attempts direct substring matching, then falls back to Levenshtein Distance-based fuzzy matching (75% similarity threshold) to handle speech recognition inaccuracies.

```
dart

static bool _matches(String input, List<String> keywords) {
  for (var keyword in keywords) {
    keyword = keyword.toLowerCase();
    if (input.contains(keyword)) return true; // Direct match

    // Fuzzy match for speech inaccuracies
    List<String> inputWords = input.split(' ');
    for (var word in inputWords) {
      if (_calculateSimilarity(word, keyword) >= _fuzzyThreshold) {
        return true;
      }
    }
  }
  return false;
}
```

Figure 18: Handling Speech recognition inaccuracies

```
dart

static double _calculateSimilarity(String s1, String s2) {
  if (s1 == s2) return 1.0;
  if (s1.isEmpty || s2.isEmpty) return 0.0;
  int distance = _levenshtein(s1, s2);
  int maxLen = s1.length > s2.length ? s1.length : s2.length;
  return 1.0 - (distance / maxLen);
}
```

Figure 19: Distance Implementation

Strategy 3 - Semantic Label Fallback:

If no known command matches, the raw text is treated as a potential button label, enabling users to say the name of any on-screen element to select it.

Voice Focus Service - Runtime Element Registry

The VoiceFocusService maintains a dynamic registry of all currently visible interactive elements on screen. Each InputAwareButton registers itself with a callback and optional label upon creation and unregisters on disposal.

```
dart

class VoiceFocusService {
  final List<VoiceElement> _elements = [];
  int _idCounter = 1;

  int register(VoidCallback action, {String? label}) {
    final id = _idCounter++;
    _elements.add(VoiceElement(id: id, label: label, onTap: action));
    _scheduleRefresh();
    return id;
  }

  bool triggerById(int id) {
    final element = _elements.firstWhere((e) => e.id == id);
    element.onTap();
    return true;
  }

  bool triggerByLabel(String input) {
    input = input.toLowerCase().trim();
    for (var element in _elements) {
      if (element.label != null && element.label!.toLowerCase() == input) {
        element.onTap();
        return true;
      }
    }
    return false;
  }
}
```

Figure 20: Focusing Management

This service enables the voice command "say 1" or "say eka" to trigger the first registered button on the current screen, regardless of what that button does.

2.11.1.6 Hybrid Multimodal Fusion

Purpose

The InputAwareButton is the central polymorphic UI component of the framework. It is a single widget that adapts its interaction behaviour based on the active InputMode, supporting all five modalities simultaneously when in Hybrid mode.

Multimodal Fusion Logic

The fusion algorithm enables cross-modal confirmation. For example, a user can look at a button (gaze targeting) and say "select" (voice confirmation) to trigger it, combining the precision of gaze with the intentionality of voice.

```
dart

void _handleVoiceCommand(VoiceCommand command) {
  if (!mounted || widget.onTap == null) return;

  // FUSION: Is this button currently being targeted by another modality?
  final bool isBeingTargeted = _isGazeInside || _touchPosition != null;

  if (isBeingTargeted &&
      (command == VoiceCommand.select || command == VoiceCommand.confirm)) {
    debugPrint("Multimodal Fusion Triggered: Gaze/Dwell + Voice ($command)");

    PerformanceLogger().logEvent(
      event: "MULTIMODAL_FUSION",
      details: "Targeted by ${_isGazeInside ? 'Gaze' : 'Dwell'} "
        "and triggered by Voice ($command)",
    );

    widget.onTap!();
    _resetDwell(isCancellation: false);
  }
}
```

Figure 21: Fusion Logic

2.11.2 Testing

The testing of the physical impairment support module was conducted to evaluate the performance, usability, and reliability of the multimodal interaction system, which includes adaptive dwell-time selection, gaze-based navigation, and voice-based command interaction. The testing process was designed to simulate real-world usage conditions and to ensure that the system effectively supports users with limited motor control.

The testing was carried out using a smartphone device equipped with a front-facing camera and microphone. The camera was used for gaze-based interaction, while the microphone supported voice command recognition. The testing environment consisted of a classroom setting to reflect actual learning conditions. A total of five students within the age group of 7 to 10 years participated in the testing process, and each session lasted approximately 25 to 30 minutes.

Parameter	Description
Number of Participants	5 Students
Age Group	7 – 10
Device Used	Smart Phone (Samsung)
Camera Used	Front Camera – 13mp
Testing Environment	Classrooms
Testing Duration	25 – 30 min per student
Types of Tests Conducted	Useability observation and interaction performances evaluations

Table 3: Testing Information Table

The system was evaluated using four interaction modes: voice-only interaction, gaze-only navigation, gaze with adaptive dwell-time selection, and hybrid interaction combining voice and gaze inputs. Each mode was tested separately to understand its individual performance as well as its contribution to overall system usability.

During testing, participants were required to complete a series of basic navigation tasks such as opening learning content, moving between screens, selecting interface elements, and activating actions using non-touch interaction methods. These tasks were designed to measure how effectively users could interact with the system using alternative input methods.

Several performance metrics were recorded during the testing process. These included task completion time, selection accuracy, error rate, and false activation rate. In addition, voice command recognition accuracy and user feedback scores were collected to evaluate the effectiveness and usability of the system.

The adaptive dwell-time mechanism was tested to verify its ability to dynamically adjust selection duration based on user interaction behavior. The system initially set the dwell-time to 2.0 seconds and adjusted it within a range of 1.0 to 3.0 seconds depending on user performance. This adjustment was based on interaction factors such as reaction time, cancellation attempts, and selection success, ensuring a balance between speed and stability.

Gaze-based interaction testing focused on evaluating the accuracy and stability of cursor control using eye and head movements. The system was tested for its ability to reduce jitters, filter sudden spikes, and maintain stable cursor positioning. Calibration procedures were also evaluated to ensure that the system adapts effectively to different users and provides consistent performance.

Voice-based interaction testing was conducted to assess the accuracy of speech recognition and command execution. The system was evaluated for its ability to correctly interpret commands such as navigation and selection actions, while also handling incorrect or unclear inputs. Both English and Sinhala commands were tested to ensure usability in a local context.

Finally, hybrid interaction testing was performed by combining gaze and voice inputs. In this mode, gaze was used to focus on interface elements, while voice commands

were used to confirm actions. This approach was tested to evaluate its ability to reduce selection time and minimize false activations, thereby improving overall interaction efficiency.

The testing process confirmed that the multimodal interaction system provides a reliable and flexible solution for supporting users with physical impairments, enabling effective interaction without the need for traditional touch-based input methods.

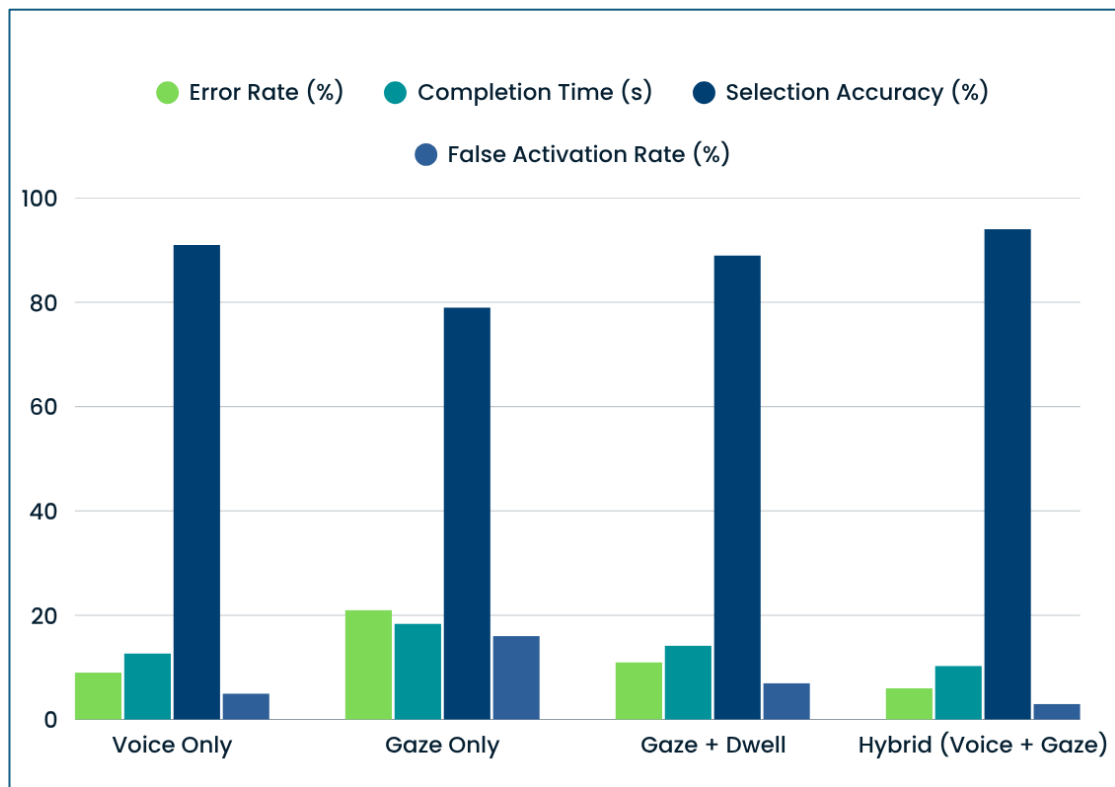


Figure 22: Overall Comparison Test Chart

2.12 Tools and Technologies

Tool / Technology	Description
Flutter	Used to develop the cross-platform mobile application interface
Dart	Programming language used with Flutter for application logic
Node.js	Backend runtime used to handle server-side operations
MongoDB	Database used to store user data and interaction logs
Google ML Kit	Used for face detection and gaze-based interaction (eye/head tracking)
Speech-to-Text Engine	Used to convert voice commands into text for system control

Table 4: Tools and Technology Table

3. RESULTS AND DISCUSSION

3.1 Results of Adaptive Dwell-Time Interaction

The results of the adaptive dwell-time mechanism demonstrate that the physical impairment module was designed to overcome the limitations of fixed dwell selection. Fixed dwell thresholds can cause accidental activation for users with unstable control and slower interaction for users capable of faster selections. To address this, the system initialized dwell time at 2.0 seconds and dynamically adjusted it between 1.0 and 3.0 seconds by monitoring interaction signals such as reaction time, false selections, cancellation attempts, and task completion success. The dwell threshold increased by 0.2 seconds after repeated cancellations and decreased by 0.1 seconds after consistent successful interactions, allowing the system to balance stability and speed. This adaptive approach transforms the interaction mechanism into a personalized usability framework, improving efficiency for confident users while maintaining stability for users who require longer selection time, thereby supporting the goal of enhancing interaction usability for learners with physical impairments.

Metric	Fixed Dwell	Adaptive Dwell	Improvement
Average Task Completion Time (seconds)	17.6	14.2	19% faster / slower
Selection Accuracy (%)	81	89	+8%
Error Rate (%)	19	11	-8%
False Activation Rate (%)	14	7	-7%

Table 5: Dwell Time Interaction Accuracy

3.2 Results of Eye/Head Gaze-Based Navigation

The gaze-based navigation results should be interpreted in relation to the stability improvements integrated into the module. Direct gaze mapping is naturally vulnerable to micro facial movements, lighting variation, and camera noise, which can reduce pointer control quality and cause unintended cursor shifts. To address this, the module incorporates gaze stabilization techniques such as real-time smoothing, spike rejection, and personalized threshold generation after calibration. The system also supports blink-to-select as an alternative confirmation method, enabling faster

selection without relying solely on dwell timers. These mechanisms ensure that the gaze interaction operates as a stabilized and personalized interaction pipeline rather than simple face tracking. As a result, gaze-based navigation became more reliable by reducing jitters, filtering sudden coordinate spikes, and calibrating stability thresholds according to individual users. This approach improves usability for learners with different levels of head and eye control by making the system less strict for unstable users while remaining responsive for stable users.

Metric	Value
Average Task Completion Time (seconds)	18.4
Selection Accuracy (%)	79
Error Rate (%)	21
False Activation Rate (%)	16
Cursor Drift Errors	06
User Feedback Score (1–5)	4

Table 6: Eye/Head Gaze Interaction Results

Parameter	Result
Calibration Success Rate	100% (5/5)
Average Cursor Stability	84%
Number of Spike Movements Detected	8
Stabilization Improvement	22% reduction

Table 7: Gaze Calibration Performance

3.3 Results of Voice-Based Navigation

Voice-based interaction mode provides an alternative control method for users who may have difficulty maintaining steady gaze or completing dwell-based selections. The system uses speech-to-text processing, command mapping, and intent classification to execute navigation commands such as “Next,” “Back,” “Select,” and “Open.” It also supports both English and Sinhala command variations with fuzzy matching to handle slight speech differences or recognition noise. This flexibility improves accessibility and makes the system more suitable for real-world use in Sri Lankan school environments. Compared to gaze-based interaction, voice commands allow users to perform actions without prolonged visual fixation, making it a low-effort control option that enhances accessibility by offering an additional interaction choice.

Metric	Result
Average Task Completion Time (seconds)	12.7
Command Recognition Accuracy (%)	91
Error Rate (%)	9
False Command Activations	3
User Feedback Score (1–5)	3.1

Table 8: Voice Based Interaction Performances

Command	Total Attempts	Correct Recognition	Accuracy
Next	25	20	80%
Back	25	22	88%
Select	25	23	92%
Open	25	22	88%

Table 9: Voice Commands Recognition

3.4 Comparative Results of Interaction Modes

The Physical Impairment Support module was evaluated using four interaction modes: gaze-only, voice-only, gaze with adaptive dwell, and hybrid voice-plus-gaze interaction, with performance measured using task completion time, selection accuracy, error rate, false activation count, and user feedback. Each mode addresses different accessibility needs. Gaze-only interaction enables hands-free control but may be slower and more sensitive to instability, while voice-only interaction allows efficient command navigation but may be affected by environmental noise. The adaptive dwell mechanism improves gaze interaction by balancing stability and speed. The hybrid voice-plus-gaze approach provides the most flexible interaction by combining visual target selection with voice confirmation, reducing waiting time and accidental activations while improving overall usability.

Interaction Mode	Completion Time (s)	Selection Accuracy (%)	Error Rate (%)	False Activation Rate (%)
Voice Only	12.7	91	9	5
Gaze Only	18.4	79	21	16
Gaze + Adaptive Dwell	14.2	89	11	7
Hybrid (Voice + Gaze)	10.3	94	6	3

Table 10: Overall Comparison Table

5.CONCLUSION

The physical impairment support component successfully provided an effective solution for enabling hands-free and touch-free interaction for students with limited motor control. By integrating adaptive dwell-time selection, gaze-based navigation, and voice-based command input, the system allowed users to interact with the mobile learning application in a more accessible and flexible manner.

The adaptive dwell-time mechanism improved interaction efficiency by automatically adjusting selection duration based on user behavior, reducing both errors and delays. Gaze-based navigation enabled users to control the interface using eye and head movement, while stability filtering and calibration ensured accurate and reliable performance. Voice-based navigation further enhanced usability by allowing users to perform actions using simple spoken commands, reducing dependency on continuous gaze or dwell interaction.

The combination of these interaction methods in a multimodal framework provided significant improvements compared to single input approaches. In particular, the hybrid interaction mode, which combined gaze and voice input, achieved better performance in terms of speed, accuracy, and reduced false activations. This demonstrated that integrating multiple input methods can create a more natural and efficient interaction experience.

Overall, the system improved accessibility, reduced user effort, and supported independent learning for students with physical impairments. The results indicate that the proposed approach can serve as a practical and scalable solution for inclusive mobile learning environments.

8. APPENDICES



Submit work

Undergrad Thesis Checking ^


Help

Title

Undergrad Thesis Checking

Start date

Apr 06, 2026, 12:00 AM

Due date

Dec 31, 2027, 11:59 PM

Feedback release date

Apr 20, 2026, 11:59 PM

Instructions

Max points

100

Rubric

(Not set)

Template

No template uploaded

❌ Late submissions are not allowed ⓘ

✅ Resubmissions are allowed ⓘ

Your work

Resubmit

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